Online Anchor-Robust Forecasting under Economic Regime Shift

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Running title: Online anchor-robust forecasting

Abstract

Forecasting models are fitted on past regimes, and they can fail when a driver such as geopolitical risk or policy uncertainty moves to a level not seen before. Most online methods respond after the shift, by detecting it and recalibrating. We study whether a forecaster can be made robust in advance when the shift moves an observed driver. OARF runs anchor regression online: it penalises the part of the forecast error that is correlated with the driver, and with a zero penalty it is online gradient descent. We prove a no-regret bound against a comparator evaluated under the worst bounded shift of the driver. In simulations where the driver causes the shift, OARF more than halves the worst-case error of online gradient descent, at some cost in ordinary error, and stress tests show when the advantage disappears. In weekly volatility forecasting for Brent crude and Henry Hub gas, every combination method beats a persistence forecast, but OARF is no better than online gradient descent or an equal-weight average.

Keywords: Online learning; Combining forecasts; Robustness; Volatility forecasting; Structural change; Causality.

JEL: C53, C22, C58, Q47.

1 Introduction

A forecasting model is estimated on data from the regimes it has seen. When a geopolitical shock, a policy turn or an energy crisis moves the economy into a new regime, relations that held in the estimation sample can break, and a model that was well calibrated last month can be badly miscalibrated this month. The usual response in online forecasting is to detect the break and adapt: monitor a loss or a coverage statistic, raise an alarm when it drifts, then recalibrate or re-weight (Gibbs and Candès, 2021; Khan and Syed, 2025; Prinster et al., 2025). This works, but the alarm can only fire after the loss has risen, so the forecaster pays a re-adaptation cost at every break.

This paper asks a narrower question. Suppose the shift moves a variable that the forecaster observes, such as an index of geopolitical risk, policy uncertainty or market-wide volatility. Can the forecaster be built in advance so that its errors do not depend on that variable, and what does this cost when no shift occurs? Anchor regression (Rothenhäusler et al., 2021) answers this question for batch data. It penalises the part of the residual that is correlated with a designated "anchor" variable, and under a linear structural model this is equivalent to minimising the worst-case error over bounded shifts of the anchor. We make the estimator online and test it on simulated and real forecasting problems.

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An example. The following example shows what changes across regimes, what stays stable, and why penalising the anchor helps. A target y, which could be next week's volatility, is driven by a stable predictor x_{p}: $y = x_{\mathrm{p}} + e_y$. A second predictor x_{c} tracks the target but is also moved by an observed driver a: $x_{\mathrm{c}} = y + a + e_c$. One can think of x_{c} as a news-based indicator that rises with volatility but also with the volume of geopolitical headlines. In the estimation sample a has two regimes, with means -1 and $+1$. In the forecast period its mean moves to $+3$, a level never seen in sample. The relation between y and x_{p} does not change; the relation between y and x_{c} shifts by the change in a. Table 1 compares least squares with anchor regression that uses a as the anchor (the script is in the replication files). Least squares puts some weight on x_{c}, because in sample it carries information about y, and it has the lower in-sample error. When a moves to the new level that weight becomes a liability, and the error of least squares roughly doubles. Anchor regression sees that the residual of a predictor that leans on x_{c} moves with a, removes that weight, and loses almost nothing in the new regime. It gives up some in-sample accuracy (0.240 against 0.206) in exchange.

Table 1: The example of Section 1. Weights and mean squared errors of least squares and of batch anchor regression with penalty ξ, all fitted on the two-regime estimation sample and evaluated on that sample and on a new regime in which the mean of the driver a moves from ± 1 to 3 (4,000 draws in each sample).

Method	weight on x_{p}	weight on x_{c}	MSE, estimation sample	MSE, new regime
least squares	0.815	0.176	0.206	0.452
anchor regression, $\xi = 10$	0.969	0.021	0.240	0.242
anchor regression, $\xi = 100$	0.986	0.003	0.249	0.247
x_{p} alone	1	0	0.250	0.248

The example also shows what the method needs. The anchor must be observed, and it must be the variable through which the shift travels. If the analyst chose a variable unrelated to the shift, the penalty would remove useful signal and buy no protection. Sections 4 and 5 measure how much of the benefit survives when the anchor is noisy, misaligned or only weakly related to the shift.

What the paper does. Section 3 turns the anchor-regression objective into a streaming first-order update, OARF (online anchor-robust forecasting), that uses exponentially weighted moving averages of the relevant moments. With the penalty set to zero it is online gradient descent, so one parameter controls how much in-sample accuracy is traded for robustness. We define an interventional regret, in which the comparator is evaluated under the worst bounded shift of the anchor, and prove that the update is no-regret in this sense when the anchor is fixed. Section 4 uses a simulated structural model in which the anchor causes the regime shifts, so that robustness can be measured on shifts held out from training, and stress-tests the method with noisy, misaligned and misspecified anchors. Section 5 applies it to weekly volatility forecasting for Brent crude and Henry Hub natural gas, with macro-uncertainty indices as the anchor. Section 6 treats the case in which the analyst does not know the anchor and describes a rule that estimates it from the data, together with a check of when that estimate can be trusted.

The findings are mixed, and we report them as they are. In simulation, where the anchor causes the shifts, OARF more than halves the worst-case error of online gradient descent, at a cost in ordinary error. The advantage disappears when the anchor is observed with noise of about one standard deviation or is misaligned with the true channel by more than about 50°, and it reverses when regimes shift variances or the target equation itself. On the energy data, every combination

method beats a persistence forecast by a wide margin, but OARF is not better than online gradient descent or an equal-weight average, and the Model Confidence Set retains almost all methods. The learned-anchor extension recovers the right subspace on most simulated streams but fails badly on a minority, and its reliability check catches most of those failures.

2 Related work

Our method draws on causal robustness, on online methods for distribution shift, and on online convex optimisation.

Causal robustness and invariance. The starting point is the observation that a predictor which is *invariant* across environments is also robust to interventions that move between them. Invariant causal prediction (Peters et al., 2016) formalises this for the identification of causal parents. Anchor regression (Rothenhäusler et al., 2021) turns it into an estimator. By penalising the projection of the residual onto an anchor variable, it interpolates between ordinary least squares and an instrumental-variables solution, and the penalised problem is provably equivalent to minimising worst-case risk over a bounded set of interventions on that anchor. DRIG (Shen et al., 2023) extends the principle to distributional robustness through invariant gradients. All of these are batch procedures that take the environment or anchor variable as given, which is the obstacle for a streaming deployment. OARF keeps the worst-case-intervention interpretation of anchor regression and makes the estimator online. We also study a heuristic add-on that estimates the anchor channel from data. Because that estimate can be unreliable, we pair it with an online diagnostic that flags streams on which the learned channel should not be used; on those streams the forecaster reverts to plain online gradient descent. Our theorem covers only the fixed, analyst-supplied channel. For a deployment that cannot supply one, the diagnostic is a partial safeguard.

Reactive online adaptation. A second, very active line of work treats distribution shift as something to detect and then accommodate. Adaptive conformal inference (Gibbs and Candès, 2021) adjusts a miscoverage level online to maintain coverage under drift. Methods from 2025 tighten the detect-then-adapt loop considerably. M-FISHER (Khan and Syed, 2025) maintains an exponential test martingale over non-conformity scores and, when the martingale crosses a Ville threshold, takes a Fisher-preconditioned natural-gradient correction. WATCH/WCTM (Prinster et al., 2025) monitors a weighted conformal test martingale that tolerates benign covariate shift while still flagging harmful concept shift, and adapts the prediction interval accordingly. These are the incumbents against which a proactive method should be judged. OARF tests whether, when an informative anchor is available, the need for reactive triggering can be partly reduced by removing the channel's influence before it raises the loss.

Distributionally robust forecasting. The closest competitors are recent methods that, like OARF, optimise an explicitly worst-case objective. Wang (2026) propose online randomised distributionally robust forecast combination for dependent data. They draw combination weights from a distribution over the simplex and update it by stochastic mirror descent on a Wasserstein ambiguity objective. Liu and Cheng (2026) develop adaptive robust combination with variance- and Expected-Shortfall-scaled exponential weights for yield-curve forecasting, and Chen et al. (2026) analyse Wasserstein distributionally robust online learning as a primal–dual game with regret guarantees. A parallel strand makes conformal decisions cost-sensitive (Wu et al., 2026). Several of

these 2025–2026 works are recent preprints, so we treat them as contemporary literature and competitive baselines, not as settled, peer-reviewed results. They differ from OARF in geometry. They hedge over an *isotropic* ambiguity ball centred at the empirical distribution, which spreads robustness equally in all directions, including directions along which the data never shift. OARF hedges over the anchor-shaped ellipsoid $\mathbb{E}[aa^\top]$, so its robustness budget is spent only along the anchor directions in which shifts occur. Our experiments are designed to isolate this difference. They show that the anchor-shaped geometry improves interventional robustness when the anchor is informative, and that the gain shrinks or reverses when it is not.

Adjacent online and subspace methods. Nearby work also informs our design. Online invariant and domain-adaptation methods seek predictors that hold across changing environments in a streaming setting. OARF can be seen as an online, single-anchor instance of this idea, specialised to mean-shift interventions, with a closed-form robustness term in place of an explicit environment partition. Nonstationary-subspace and change-adaptive analysis (subspace tracking, stationary-subspace analysis, and the path-length-adaptive online learning of Zhang et al., 2018) is what our discovery layer does: it is an unsupervised nonstationary-subspace estimator, and for that reason we do not call its output causal. What separates OARF from these methods is that it feeds the discovered nonstationary subspace back into an anchor-regression objective whose target is interventional, not purely predictive.

Online convex optimisation. The regret analysis rests on dynamic-regret results for online gradient descent (Zinkevich, 2003; Hazan, 2016), the path-length-adaptive refinement of Zhang et al. (2018), and AdaGrad-style preconditioning (Duchi et al., 2011). The economic measurements draw on the daily Geopolitical Risk index (Caldara and Iacoviello, 2022), the Economic Policy Uncertainty index (Baker et al., 2016), realised-volatility modelling in the heterogeneous-autoregressive tradition (Corsi, 2009), and the volatility-timing logic of Fleming et al. (2001).

3 The OARF method and its guarantee

Setup and protocol. We work in discrete time $t = 1, \dots, T$. At each step the forecaster observes predictors $x_t \in \mathbb{R}^d$ and a vector of candidate regime drivers $z_t \in \mathbb{R}^p$, and must predict a scalar target y_t. In a pure forecasting problem the predictors are explanatory variables; in a combination problem they are the forecasts of several base models. A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w emits $\hat{y}_t = w^\top \phi(x_t)$, incurs the squared loss $\ell_t(w) = (y_t - \hat{y}_t)^2$, and writes its residual $r_t = y_t - \hat{y}_t$. The *anchor* is a (possibly learned) linear read-out of the context, $a_t = B^\top z_t \in \mathbb{R}^q$, with channel matrix $B \in \mathbb{R}^{p \times q}$. The learning protocol is sequential. At step t we first predict from x_t, then reveal y_t and update, and each learner standardises its inputs with past-only statistics. When we later evaluate a frozen learner on held-out interventions, we replay the same frozen standardiser.

Objective. Following anchor regression (Rothenhäusler et al., 2021), OARF trades predictive fit against invariance of the residual to the anchor by minimising

$$b_{\mathrm{AR}}(w; \xi) = \underbrace{\mathbb{E}\big[(y - w^\top \phi(x))^2\big]}_{\text{predictive loss}} + \xi \underbrace{\mathbb{E}[ar]^\top \big(\mathbb{E}[aa^\top]\big)^{-1} \mathbb{E}[ar]}_{P(w) = \| \operatorname{proj}_a r \|^2}, \qquad \xi \geq 0. \tag{1}$$

The penalty $P(w)$ is the squared length of the projection of the residual onto the anchor. Driving it down forces $\mathbb{E}[ar] \to 0$, so the predictor stops using any component of the signal that co-moves with

the regime drivers. At $\xi = 0$ the objective is ordinary least squares, and as ξ grows the solution moves toward the invariant predictor. The objective also has a robustness interpretation.

Proposition 1 (Interventional-robustness equivalence, Rothenhäusler et al., 2021, Thm. 1)**.** *Assume a linear structural causal model in which (a) the anchor A is a* source *(exogenous) node; (b) the remaining variables, including the target, follow linear structural assignments $V = \mathbf{B}V + \varepsilon + \mathbf{M}A$ with $A \perp \varepsilon$ and $I - \mathbf{B}$ invertible; (c) the considered interventions are mean-shift interventions* $\mathrm{do}(a := \nu)$ *that replace A by a constant; and (d) the loss is squared error. Then for every $\xi \geq 0$ there is a radius $\rho(\xi)$, given in closed form by the cited theorem and strictly increasing in ξ (with $\rho(0) = 0$, the OLS support, and $\rho(\xi) \to \infty$ as $\xi \to \infty$, the instrumental-variables limit), such that*

$$\min_w b_{\mathrm{AR}}(w; \xi) \;=\; \min_w \sup_{\nu:\, \|\nu\| \leq \rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}\big[(y - w^\top \phi(x))^2\big]. \tag{2}$$

The equivalence does not hold for arbitrary nonstationarity. It requires the exogenous-anchor SCM (a)–(c) above, and the worst case is taken over mean-shift interventions only. Under these conditions the set hedged over is an ellipsoid shaped by the anchor covariance $\mathbb{E}[aa^\top]$, whereas the distributionally robust methods of Section 2 hedge over an isotropic ball; our evaluation tests whether this difference in geometry matters. Our theory inherits the same assumptions (Assumption 2). Where they fail, the stress tests of Section 4 measure the degradation.

Streaming update. The gradient of (1) in the coefficient block separates into a prediction term and a robustness term,

$$\nabla_w b_{\mathrm{AR}} = -2\,\mathbb{E}[\phi r] \;-\; 2\xi\,\mathbb{E}[\phi a^\top]\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar], \tag{3}$$

and to move from batch to streaming we replace each population moment by an exponential moving average (decay β) of its sample analogue, computed on centred variables. Subtracting the running means removes the across-regime mean component of the anchor. Under the SCM of Proposition 1, where a regime is a mean shift on A, this is the component that varies between regimes, and what remains is the within-regime second-moment structure used by the penalty in (1). Centring does not remove the intervention itself, whose effect persists in X and Y. It only keeps the slowly varying regime mean out of the covariance estimate. The channel-discovery rule of Section 6 uses the across-regime mean variation that centring discards. Writing $\tilde{\cdot}$ for the EMA-centred quantities, and collecting the non-intercept coefficients as w_{-0}, we maintain running estimates m_t^{Xr}, m_t^{XA}, M_t^{AA}, m_t^{Ar} of $\mathbb{E}[\tilde{\phi}\,\tilde{r}]$, $\mathbb{E}[\tilde{\phi}\,\tilde{a}^\top]$, $\mathbb{E}[\tilde{a}\tilde{a}^\top]$, $\mathbb{E}[\tilde{a}\tilde{r}]$ (here $\tilde{\phi}$ stacks the intercept and centred $\tilde{x}$, and the intercept row of m_t^{XA} is held at zero so the penalty never acts on the intercept), and update

$$\boxed{w_{t+1} = w_t - \eta\, g_t, \qquad g_t = \underbrace{-\,r_t\,\phi(x_t)}_{\text{prediction}} \underbrace{-\,2\xi\,[\,0;\, m_t^{XA}(M_t^{AA} + \epsilon I)^{-1} m_t^{Ar}\,]}_{\text{anchor robustness}} + \lambda_2[0; w_{t,-0}].} \tag{4}$$

Of the two factors of -2 in (3), the prediction one is absorbed into the step size η and the robustness one is kept explicit, so that ξ reads directly as a relative weight. Signs follow gradient descent: the ridge term $+\lambda_2[0; w_{-0}]$ enters g_t with a positive sign, and the update $w \leftarrow w - \eta g_t$ therefore shrinks the non-intercept weights. The ridge ϵ stabilises the anchor Gram matrix M_t^{AA} when it is ill-conditioned (we use $\epsilon = 10^{-3}$). Each step costs $O(d{+}pq{+}q^3)$, dominated by the $q \times q$ solve, which is negligible for the small q we consider. In practice we take the step with AdaGrad per-coordinate scaling (Duchi et al., 2011). The gradient g_t is exactly (4) and only its step length is preconditioned, so the online-convex-optimisation regret machinery is unchanged, while a single global rate adapts to features on very different scales. With $\xi = 0$, g_t reduces to the plain squared-loss gradient and the update is online gradient descent; this holds regardless of AdaGrad and standardisation, which act identically on all methods. The decay β controls how quickly the moment estimates forget.

Algorithm 1 OARF, one streaming pass. The block for learning the channel is the extension of Section 6 and is skipped when the channel is supplied.

input: penalty ξ, rates η, η_B, decay β, ridge ϵ, λ_2, block L, channel B (or random init if learning)
$w \leftarrow 0$; EMA means and cross-moments $\leftarrow 0$; block buffer $\leftarrow 0$
for $t = 1, \dots, T$ **do**
    observe x_t, z_t; standardise (past-only, winsorised): x_s, z_s; $a \leftarrow B^\top z_s$
    $\hat{y}_t \leftarrow w^\top [1; x_s]$; **emit** $\hat{y}_t$; reveal y_t; $r \leftarrow y_t - \hat{y}_t$
    update EMA means; centre $\tilde{x}, \tilde{a}, \tilde{r}$; update $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$
    intercept gradient $g_0 \leftarrow -r$
    coef. gradient $g_{-0} \leftarrow -r\, x_s - 2\xi\, m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} + \lambda_2\, w_{-0}$
    $w \leftarrow \Pi_{\mathcal{W}}\big(w - \textsc{AdaGradStep}(g)\big)$ ▷ $g = [g_0; g_{-0}]$; ridge sign shrinks w_{-0}
    if learning the channel **then**
        accumulate z_s into the block buffer; **every** L steps update across-block scatter S, mean u, and set $B \leftarrow \mathrm{orth}\big(B + \eta_B\, 2(S - \gamma_c u u^\top)B\big)$
    end if
    absorb (x_t, z_t) into the standardiser statistics
end for

Dimensions: $w \in \mathbb{R}^{d+1}$ (w_0 intercept, $w_{-0} \in \mathbb{R}^d$); $m^{Xr}, x_s \in \mathbb{R}^d$; $m^{XA} \in \mathbb{R}^{d \times q}$ (intercept row excluded); $M^{AA} \in \mathbb{R}^{q \times q}$; $m^{Ar} \in \mathbb{R}^q$; channel $B \in \mathbb{R}^{p \times q}$. $\Pi_{\mathcal{W}}$ projects onto the radius-D ball.

Interventional regret. To state the guarantee we need a regret notion that rewards robustness to future interventions as well as tracking of the past. For a radius $\rho \geq 0$ define the *interventional regret*

$$\mathrm{IntReg}_T(\rho) = \sum_{t=1}^{T} \ell_t(w_t) - \min_{u \in \mathcal{W}} \sum_{t=1}^{T} \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\mathrm{do}(a:=\nu_t)}\big[(y_t - u_t^\top \phi(x_t))^2\big] : \tag{5}$$

the learner's realised loss, measured against the best comparator in a class $\mathcal{W}$ that is itself stress-tested by the worst bounded intervention at every step.

Choice of regret notion. Ordinary dynamic regret measures the learner against the best predictor on the realised sequence. It rewards fitting whatever just happened, which is the reactive behaviour we want to avoid, and says nothing about robustness to a shift that has not yet occurred. Suppose two regimes differ only in the anchor's location. A predictor that exploits an anchor-correlated signal can have low dynamic regret, because it tracks each realised regime well, and still incur a large loss as soon as the anchor moves to a new value. Interventional regret requires the comparator to do well under the worst bounded intervention, so the learner can compete only by being robust across the whole intervention class. The comparator is stress-tested while the learner is scored on realised loss. This asymmetry favours the comparator and makes the bound harder to attain. The realised stream is itself one admissible intervention (Assumption 2), so the learner is implicitly stressed by the shifts it actually meets. One could instead score learner and comparator under the same counterfactual intervention path. We do not, because the learner makes its online decisions on the realised stream and the counterfactual grid is used only for evaluation. Equation (5) is therefore a *robust-comparator* regret (realised loss for the learner against a worst-case-stressed comparator), not a fully counterfactual online regret in which the learner also receives counterfactual feedback. The guarantee does not imply superiority under arbitrary adversarial shifts, since the comparator class is regime-conditional. It gives no guarantee for the

learned channel, and it does not imply low realised MSE when the anchor is wrong; the stress tests and scenarios show that case explicitly.

The guarantee rests on the following assumptions.

Assumption 1. *(Boundedness.) The standardised, winsorised features and anchor are bounded, so the per-step gradients of the surrogate below are bounded in norm by a constant G, and the comparator class $\mathcal{W}$ lies in a Euclidean ball of radius D.*

Assumption 2. *(Linear-SCM anchor model.) The data follow a linear structural causal model in which the anchor enters as in Section 3, so the equivalence of Proposition 1 holds within each regime, and the radius is adequate, $\rho(\xi) \geq \max_t \|\nu_t^\star\|$, where $\nu_t^\star$ is the realised regime's anchor displacement (so each realised regime is one admissible intervention in the ball).*

Assumption 3. *(Oblivious regime schedule and comparator.) The regime schedule is exogenous and oblivious to the learner; the per-step worst-case risk $\mathcal{R}_t$ defined below is therefore a deterministic function of w, and $\mathcal{W}$ is the class of regime-conditional comparators that switch at most K times (path length $P_T \leq 2DK$).*

Assumption 4. *(Within-regime stationarity.) Within each regime the joint distribution of (ϕ, a, y) is stationary with mixing fast relative to the EMA window, so the EMA moment estimates obey the concentration of Lemma 3.*

We view Algorithm 1 as online gradient descent fed inexact gradients on the convex per-step worst-case risks $\mathcal{R}_t(w) := \sup_{\|\nu\| \leq \rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}[(y_t - w^\top \phi(x_t))^2]$ of (5). The full argument, in three lemmas, is given in Appendix A; here we state the result and sketch the proof.

Theorem 1 (Fixed-anchor OARF interventional regret)**.** *Under Assumptions 1–4, Algorithm 1 with a fixed channel and the Ader step wrapper attains sublinear expected interventional regret against any K-switch comparator, with no prior knowledge of K:*

(i) with oracle anchor second moments, $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big(\sqrt{T(1+K)}\big)$;

(ii) with the EMA moment estimates of Algorithm 1 and decay $1-\beta_T = T^{-2/3}$, $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big((1+K)\,T^{2/3}\big)$.

Proof sketch (full proof in Appendix A). (1) $\mathcal{R}_t$ is convex and, by Proposition 1, equals the anchor-penalised risk within a regime; the update direction is $\nabla \mathcal{R}_t(w_t) + \zeta_t$, where ζ_t splits into a zero-mean martingale part M_t (one-sample gradient noise) and a deterministic part b_t (EMA moment-estimation error). (2) The path-length-adaptive Ader wrapper (Zhang et al., 2018) gives dynamic regret $O(\sqrt{T(1+K)})$ against a K-switch comparator without knowing K, plus the inexactness term $\sum_t \langle \zeta_t, w_t - u_t \rangle$. (3) Against the deterministic (oblivious-schedule) comparator, $\mathbb{E}\sum_t \langle M_t, w_t - u_t \rangle = 0$, while $D\sum_t \mathbb{E}\|b_t\| \leq D(c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta))$ is zero with oracle moments and $O((1+K)T^{2/3})$ at $1-\beta_T = T^{-2/3}$. Bounding realised loss by the worst-case risk, $\mathbb{E}_t[\ell_t(w_t)] \leq \mathcal{R}_t(w_t)$ (Assumption 2), gives the two rates; both are $o(T)$. □

Remark 1 (Reading the two rates)*.* The gap between (i) and (ii) is the cost of learning the robustness geometry online without changepoint detection. A detection-and-reset estimator could reduce the moment error at the $O(1/\sqrt{n})$ within-regime rate and recover (i), but it would reintroduce the triggers OARF is designed to avoid. The $T^{2/3}$ rate is what a proactive, exponentially forgetting estimator pays, and it is still sublinear. The theorem applies to the Ader-wrapped version, while the experiments use a single AdaGrad instance, so the algorithm we run is not exactly the one we analyse. AdaGrad keeps per-coordinate adaptive steps but drops the K-adaptivity of the Ader

wrapper, which is what makes the bound hold for an unknown number of switches. We use it because it is simpler, has fewer tuning parameters and is more robust on heavy-tailed streams. The cost is that the worst-case K-switch guarantee is established only for the Ader variant. In a pilot on the synthetic benchmark (Appendix E, 12 seeds) the two step rules give qualitatively similar results. Both cut the worst-case interventional error far below OGD, their per-seed worst-case errors are strongly rank-correlated (Spearman $\rho = 0.80$), and AdaGrad is slightly more accurate and stable. We therefore report AdaGrad as a simpler stand-in.

Remark 2 (Scope of the guarantee). First, risk-adjusted no-regret is attainable only under stochastic, stationary or ergodic data, and not against a fully adversarial environment (Uziel and El-Yaniv, 2018). For this reason $\mathcal{W}$ is regime-conditional, and observable regime drivers must enter the statement, since they define the regimes the comparator may condition on. Second, anchor-based methods deliver interventional robustness, not causal identification. The estimand is the diluted-causal parameter, and we do not claim to recover a structural coefficient.

4 Simulation study

The simulation is built so that the anchor causes the regime shifts and so that robustness can be measured on shifts that were never seen in training. It shows what OARF can deliver when its assumptions hold and how quickly the benefit disappears as they are relaxed.

4.1 Design

Synthetic anchor-SCM. The synthetic generator (Table 2) is built so that the gap between ordinary prediction and interventional robustness can be measured. An exogenous anchor A and a hidden confounder H drive a block of causal parents X_{par} of the target, which is generated as $Y = X_{\text{par}}^{\top} b_{\text{par}} + H^{\top} d_H + \varepsilon_Y$. The mean of A shifts across eight regimes, and these shifts are the in-sample interventions. A second block of children is generated downstream of the target, $X_{\text{ch}} = Y c_{\text{ch}} + A W_{Ac} + H W_{Hc} + \varepsilon_{\text{ch}}$, so the children are anti-causal predictors that also load on the anchor. In sample the children are highly informative about Y, since they are generated from it, and ordinary least squares and online gradient descent put heavy weight on them. A child's relationship to Y is, however, contaminated by the anchor term AW_{Ac}. Under an intervention $\text{do}(A := \nu)$ this term is set externally, the child becomes misleading, and the error it injects grows with $\|\nu\|$. Only a predictor that uses the parents alone is unaffected by the intervention, and this is the gap that anchor robustness is meant to close. To give the discovery layer something to recover, the observable context z is a random rotation of the anchor together with $p - q$ stationary AR(1) decoy drivers, so there is a ground-truth channel B_{true} with $ZB_{\text{true}} = A$. We hold out a fixed grid of interventions (ten magnitudes, six directions, four hundred draws each) that extends to anchor values beyond the training range. Every interventional number in this section is computed on this grid.

4.2 Methods and evaluation

Baselines. We compare against eleven methods, grouped by the question each answers about OARF (Table 3). The *basic baselines* are online gradient descent (which is OARF with $\xi = 0$), a rolling-window ridge regression, and adaptive conformal inference (Gibbs and Candès, 2021). They show the accuracy attainable with no robustness mechanism and with a strong batch refit, and so calibrate both ends of the scale. The 2025 reactive methods, M-FISHER (Khan and Syed, 2025) and WATCH/WCTM (Prinster et al., 2025), follow the detect-then-adapt approach that OARF

Table 2: Configuration of the synthetic anchor-SCM. Unless otherwise stated, the main performance summary (Table 5) and the channel-discovery figure aggregate over 50 seeds; the component ablations (Table 6) use 16 seeds and the assumption-violation and selector experiments (Tables 7–10) use 8 seeds. Each table and figure states its own seed count.

Symbol	Meaning	Value
T	stream length	8000
$d_{\text{par}}, d_{\text{ch}}$	causal parents / anti-causal children	$3, 3$
p	observable context dimension	8
q	true anchor (channel) dimension	2
h	hidden-confounder dimension	2
#regimes	anchor-mean regimes ($\Rightarrow$ 7 changepoints)	8
σ_{shift}	across-regime anchor-mean scale	1.5
σ_A	within-regime anchor sd	0.6
c_{ch} scale	strength of the target$\to$child edge	1.3
W_{Ac} scale	anchor$\to$child contamination	1.1
confounding	strength of the $H \to (X, Y)$ path	0.9
σ_X, σ_Y	predictor / target noise	$0.4, 0.5$

is designed to replace. Comparing against them tests whether removing a channel's influence in advance does better than reacting to it after the loss rises. The 2026 distributionally robust methods are randomised Wasserstein combination (Wang, 2026), the variance- and ES-scaled exponential-weight combiners (Liu and Cheng, 2026), Wasserstein online learning (Chen et al., 2026) and cost-aware conformal inference (Wu et al., 2026). They are the closest comparison, because they also optimise a worst-case objective. They differ from OARF mainly in the shape of the set they hedge over, an agnostic ball in place of an anchor-shaped ellipsoid, so this comparison bears on the value of the anchor geometry. Finally, the batch causal-robust estimators, anchor regression (Rothenhäusler et al., 2021) and DRIG (Shen et al., 2023), use the same invariance principle as OARF, but offline and with the channel supplied. They are offline references. Anchor regression, which is given the true anchor, is the natural best case for the fixed-channel online method. All methods are run under the same protocol, with AdaGrad-stabilised steps where applicable. Appendix D gives, for each baseline, the loss it optimises, its update rule, its output type, whether it is online, rolling or batch, and whether it uses the anchor.

Forecast output. Every method produces a one-step point forecast and is scored by the same squared loss under the same protocol. The linear learners (OARF, OARF-CD, OGD, Rolling-OLS, W-DRO-OL) and the combination methods (DR-Combo, FC-DRO/-ES) output $w_t^\top \phi(x_t)$ directly. The conformal methods use a quantile read-out. ACI and CostACI maintain 5/50/95% pinball heads and take the median head as the point forecast (their adaptive level affects only the interval), and WATCH/WCTM uses an OGD point head with a martingale-driven interval. Only these three also report coverage, CRPS and width. M-FISHER uses its adapted point head. All methods are therefore comparable on point loss, and the interval-based ones are also compared on distributional metrics where these are defined.

Metrics and protocol. We report point accuracy (MSE overall, in each regime, in the worst regime, and in the 50 periods after each regime change), interventional robustness on the held-out grid (the curve MSE(ν) and its maximum), and coverage, CRPS and interval width for the methods that produce intervals. Significance is assessed with the pairwise Diebold–Mariano test (Diebold

Table 3: The eleven comparison methods, organised by the question each answers about OARF. The column "Hedges over" names the uncertainty geometry; OARF is the only method whose set is an anisotropic anchor-shaped ellipsoid, and the only proactive (non-triggered) robust learner.

Method	Year	Family	Hedges over	Posture
OARF (+ OARF-CD, expl.)	ours	anchor-robust OCO	anchor ellipsoid $\mathbb{E}[aa^\top]$	proactive
OGD	2003	basic baseline	–	–
Rolling-OLS	–	basic baseline	–	windowed
ACI	2021	conformal	coverage level	reactive
M-FISHER	2025	test-time adaptation	– (martingale trigger)	reactive
WATCH/WCTM	2025	conformal martingale	interval width	reactive
DR-Combo	2026	robust combination	Wasserstein ball	mirror descent
FC-DRO(-ES)	2026	robust combination	variance / ES	exponential weights
W-DRO-OL	2026	robust online learning	Wasserstein ball	primal–dual
CostACI	2026	cost-aware conformal	coverage level	reactive
AnchorReg / DRIG	'21/'23	causal (batch)	anchor / environments	windowed

and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011). The first fifth of each stream is a warm-up, used to initialise the past-only standardisers and to fix hyper-parameters once; the rest is evaluated. The main synthetic summary aggregates 50 seeds (medians with interquartile ranges), the ablations 16 seeds and the scenario and selector experiments 8 seeds, so the absolute error of a method differs between tables and comparisons are meant within a table. All hyper-parameters are listed in Table 4 and were fixed in advance or set once on the warm-up block. The held-out intervention grid is used only for evaluation. Parts of the analysis code were written with the help of an AI coding assistant (see the acknowledgments); all of it is public and regenerates every reported number.

Table 4: Hyper-parameters. Values were fixed a priori or chosen once on the warm-up window; none were tuned on the evaluation period or the intervention grid.

Symbol	Meaning	Value
η	learning rate (AdaGrad base)	0.1
ξ	robustness penalty (headline OARF)	6 (synthetic), 1.5 (real)
β	EMA decay for moments	0.98
ϵ	ridge on the anchor Gram	10^{-3}
λ_2	ℓ_2 weight	10^{-4}
η_B	channel-discovery rate	0.3
L	discovery block length	40
γ_c	grand-mean penalty (discovery)	0.3
q	channel dimension	2 (synthetic), 3 (real)
winsorisation	standardised-feature clip	± 8
α	nominal miscoverage (conformal heads)	0.1
warm-up	fraction reserved for warm-up/HP	0.2

4.3 Results

Synthetic benchmark. Table 5 and Figures 1–4 report the controlled benchmark over 50 seeds, as medians with interquartile ranges. The in-distribution and interventional rankings are nearly reversed. In distribution, where every regime has been seen, the batch and reactive methods are the

most accurate: rolling-window OLS, DRIG and the conformal heads have the lowest median overall MSE (≈ 0.011). Fixed-channel OARF does not use the anchor-correlated child signal and pays for this with a higher median in-distribution error (0.019). Under intervention the order reverses. OARF attains a median worst-case interventional MSE of 0.041, about 57% below online gradient descent (0.095), 58% below the Wasserstein-DRO online learner (0.098) and WATCH/WCTM (0.099), and 80% below the reactive M-FISHER (0.204). It comes close to the batch anchor-regression reference (0.026) without matching it, and that reference is given the true anchor. The interquartile ranges overlap because the interventional metric is heavy-tailed and seed-dependent, so we read the gap as a robust shift in the median and not as dominance on every seed.

As the intervention magnitude grows, OARF and the batch reference stay nearly flat while the reactive and agnostic-robust methods climb steeply (Figure 3a). Sweeping ξ traces the trade-off between adaptation and robustness (Figure 3b). The update acts directly on the anchor–residual correlation: OARF holds it near zero, whereas for OGD it spikes at every regime change (Figure 4).

Table 5: Synthetic anchor-SCM over 50 seeds (lower is better). For the overall and worst-do(A) columns we report the median with the interquartile range $[\,q_{25}, q_{75}\,]$ in brackets; the worst-regime and post-changepoint columns give the median only. Medians and IQRs are robust to the heavy right tail of the interventional metric. The final column is the held-out interventional metric, and the best fully online method is in bold. AnchorReg and DRIG are batch methods, and AnchorReg is also given the true anchor, so both serve as references rather than competitors. The interventional median of OARF-CD is competitive but its tail is heavy (mean 0.147); see Section 6.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (main, fixed channel)	0.019 [0.010, 0.043]	0.030	0.021	**0.041** [**0.015**, **0.071**]
OARF-CD (*exploratory*, learned B)	0.018 [0.010, 0.040]	0.029	0.021	0.049 [0.020, 0.124]
OGD	0.012 [0.008, 0.033]	0.022	0.018	0.095 [0.031, 0.220]
Rolling-OLS	0.010 [0.005, 0.030]	0.015	0.012	0.083 [0.025, 0.278]
ACI	0.011 [0.006, 0.030]	0.015	0.013	0.071 [0.022, 0.164]
M-FISHER (2025)	0.021 [0.016, 0.043]	0.043	0.039	0.204 [0.072, 0.387]
WATCH/WCTM (2025)	0.012 [0.008, 0.033]	0.022	0.018	0.099 [0.031, 0.220]
W-DRO-OL (Chen 2026)	0.012 [0.007, 0.033]	0.022	0.018	0.098 [0.033, 0.288]
CostACI (2026)	0.011 [0.006, 0.030]	0.015	0.013	0.071 [0.022, 0.164]
AnchorReg (batch, true anchor)	0.011 [0.005, 0.031]	0.014	0.011	0.026 [0.010, 0.074]
DRIG (batch)	0.010 [0.005, 0.030]	0.015	0.014	0.116 [0.032, 0.281]

Ablations. Table 6 and Figure 5 remove one component at a time. Most of the interventional robustness comes from the anchor channel. With the true channel the worst-case interventional error is 0.080. With a random channel it rises to 0.377, which is worse than using no anchor penalty at all (OGD, 0.226). A wrong channel therefore does active harm, so identifying the right channel is the main problem; the stress tests below measure the same effect continuously. The learned channel (0.303, alignment 0.84) is worse than OGD on average in this experiment and, as Section 6 shows, unstable across seeds. AdaGrad matters for both accuracy and robustness: a tuned fixed step raises in-regime MSE from 0.035 to 0.095 and worst-do(A) from 0.080 to 0.145. EMA centring improves in-regime accuracy (0.035 vs. 0.043) with essentially unchanged robustness, which is consistent with separating the regime mean from the covariance the penalty uses. For the discovered dimension, worst-do(A) MSE and alignment both improve gradually as q rises from 1 to 4 ($0.329 \to 0.261$, alignment $0.79 \to 0.90$), because a larger learned subspace still contains the true two-dimensional channel. Under-specifying the dimension ($q = 1$) does more damage, because the smaller subspace

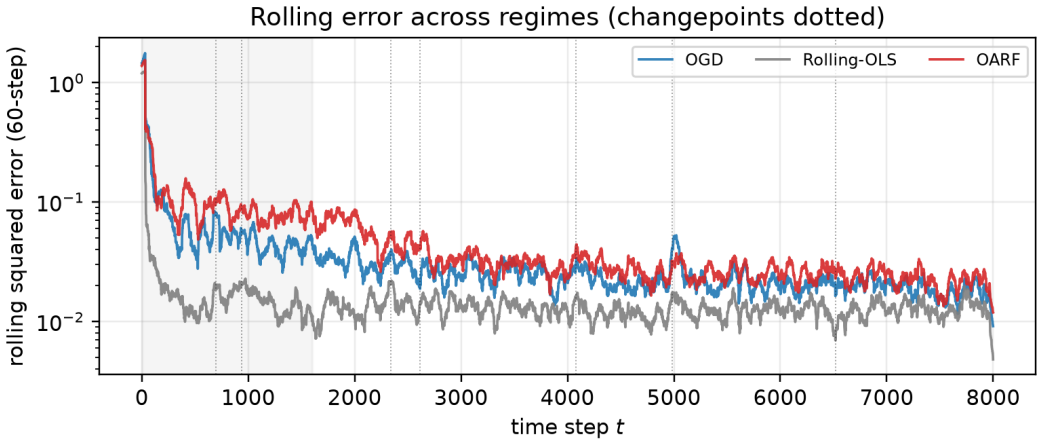


Figure 1: Rolling squared error across regimes on a representative seed (changepoints dotted, the shaded band is the warm-up window). The reactive baselines spike at every break; OARF stays comparatively flat.

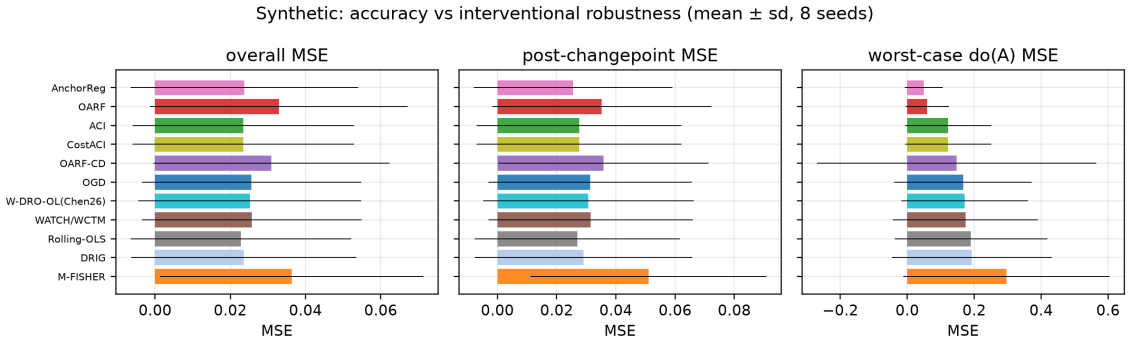
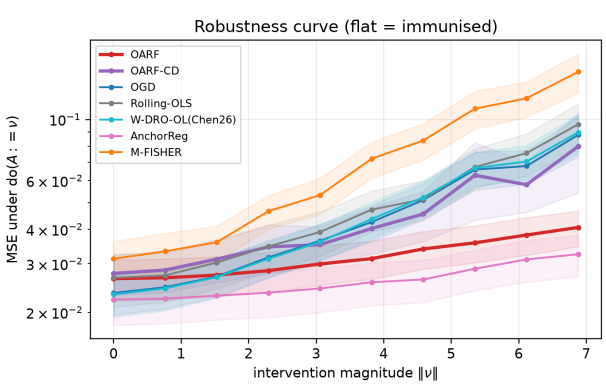
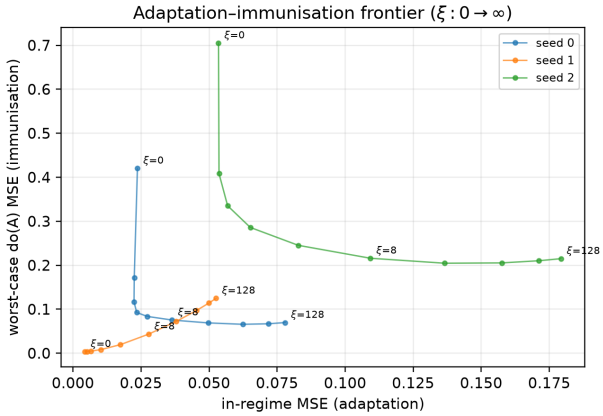


Figure 2: Overall, post-changepoint and worst-case interventional MSE on the synthetic benchmark, sorted by interventional robustness. The reactive and batch methods win in distribution and OARF and the batch oracle win under intervention, so the two panels are nearly mirror images.



(a) Robustness curve: MSE against intervention magnitude $\|\nu\|$.



(b) Adaptation–robustness frontier as $\xi : 0 \to \infty$.

Figure 3: (a) OARF and the batch oracle remain near-flat as interventions grow, while reactive and agnostic-robust methods degrade steeply. (b) The penalty ξ moves the learner continuously from the online-gradient-descent corner to maximal robustness.

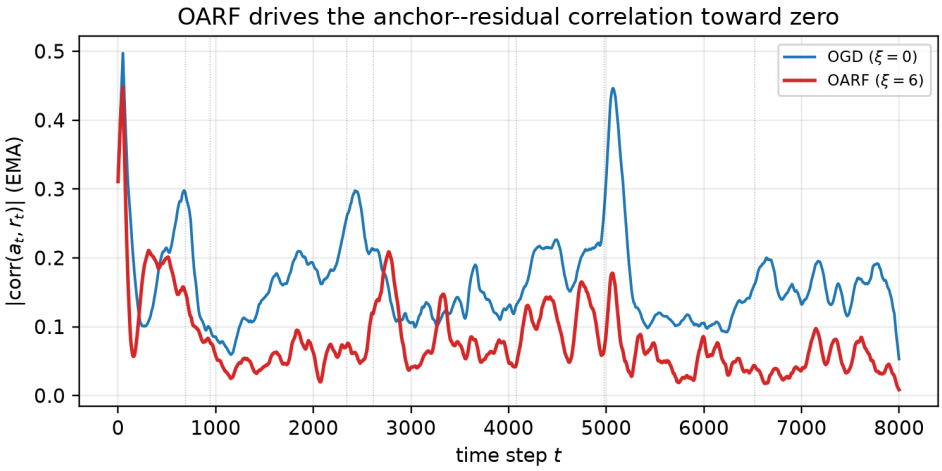


Figure 4: Anchor–residual correlation. The smoothed correlation $|\mathrm{corr}(a_t, r_t)|$ is held near zero by OARF ($\xi = 6$) but spikes for online gradient descent ($\xi = 0$) at each regime change (dotted).

cannot contain it. Results are broadly insensitive to the EMA decay β over $[0.90, 0.995]$.

Table 6: Component ablations on the synthetic benchmark (mean over 16 seeds, $\xi = 6$ unless noted). The channel carries most of the interventional robustness, and a wrong channel is worse than none.

Variant	in-regime MSE	worst-do(A) MSE
OARF (full, true channel)	0.035	**0.080**
− EMA centring	0.043	0.081
− AdaGrad (fixed step)	0.095	0.145
random channel	0.032	0.377
$\xi = 0$ (OGD)	0.027	0.226
learned channel (OARF-CD, alignment 0.84)	0.033	0.303

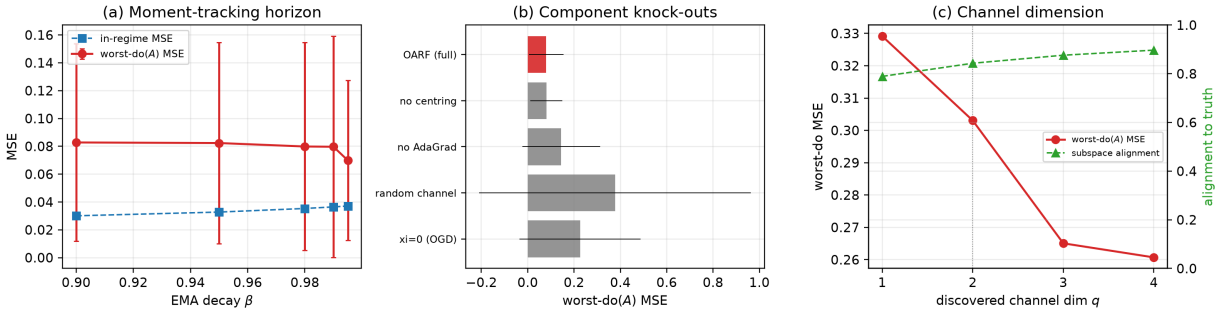


Figure 5: Ablations. (a) Sensitivity to the EMA decay β. (b) Component knock-outs by worst-case interventional MSE: the channel and the adaptive step matter most, and a random channel is worse than none. (c) Discovered-channel dimension: robustness and alignment both improve slightly with q (true $q = 2$ dotted), but worst-case error stays above OGD at every q.

Stress tests. The synthetic benchmark favours OARF by construction, because the anchor is clean and fully observed. In practice it may be neither, so we stress the fixed-channel method along two axes that break these conditions (Figure 6; the synthetic SCM is otherwise unchanged, 8 seeds). First, we add noise of growing scale to the observed context before any method sees it. Worst-do(A) MSE rises monotonically with the noise scale, and OARF crosses the no-anchor OGD baseline at roughly one standard deviation of observation noise (0.18 vs. OGD 0.19). Beyond that point a noisy anchor is no better than no anchor, and the learned channel degrades sooner. Second, we rotate the supplied channel away from the truth by an angle θ (subspace alignment $\cos\theta$). OARF beats OGD while the alignment exceeds about 0.6 (roughly $\theta \leq 50°$). With a more misaligned channel it is worse than OGD, reaching 0.36 against OGD's 0.19 when the channel is orthogonal; the random-channel row of Table 6 reflects the same mechanism. OARF should therefore be deployed only when the analyst has reason to believe the anchor is informative, that is, with alignment well above 0.6 and noise below order one. A weak or guessed channel can make forecasts worse than using no anchor.

Assumption-violation scenarios. The stress tests vary the quality of the anchor. We also change the SCM structure so as to break the assumptions behind Proposition 1 one at a time

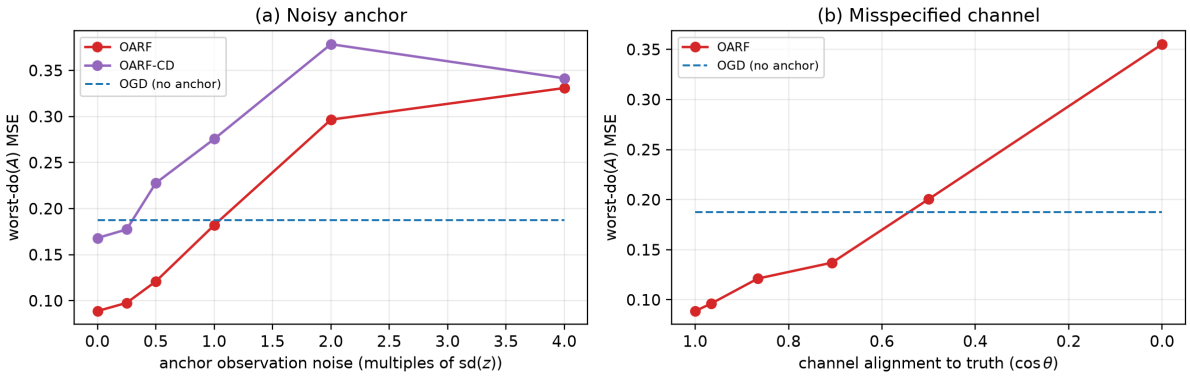


Figure 6: Stress tests of the fixed-channel method (worst-do(A) MSE; OGD baseline dashed). (a) Increasing anchor observation noise: OARF crosses the no-anchor baseline near one standard deviation of noise. (b) Rotating the channel away from the truth: anchoring helps while alignment $\cos\theta \gtrsim 0.6$ and hurts beyond that point.

(Table 7, median worst-do(A) MSE over 8 seeds). OARF keeps its advantage over OGD in three of these scenarios: when the intervention also shifts the parents (PARENTS), because the invariant predictor remains valid; when a second regime driver is unobserved (UNOBSERVED), because robustifying the observed part still helps; and when the observed anchor is a noisy proxy for the true latent driver (PROXY). The method therefore degrades gradually under these violations. It fails in two cases. When the regime shifts the target's own parent-coefficient (MECHANISM), no fixed predictor is robust and both methods collapse (≥ 1.0). When regimes shift the variance instead of the mean (VARIANCE), the mean-shift equivalence does not apply and OARF is worse than OGD. The theory covers neither case, and we regard both as limitations of the method.

Table 7: Assumption-violation scenarios (median worst-do(A) MSE over 8 seeds; lower is better). OARF uses the true observed channel. It fails under a shifting target mechanism and when regimes shift the variance instead of the mean.

Scenario	OARF	OARF-CD	OGD	assumption broken
clean (ideal)	0.021	0.033	0.051	none (reference)
parents shifted	0.018	0.016	0.047	none (still anchor-mediated)
unobserved driver	0.024	0.040	0.048	channel incompletely observed
noisy proxy anchor	0.044	0.058	0.065	anchor predictive, not causal
mechanism shift	1.299	1.238	1.020	target mechanism non-invariant
variance shift	0.046	0.040	0.025	regimes shift variance, not mean

Significance. On in-distribution loss the Model Confidence Set (computed per seed over 16 seeds) retains the batch and conformal methods (Rolling-OLS in all 16 seeds, DRIG in 15, the conformal heads in 11) and excludes fixed-channel OARF (retained in no seed) and OARF-CD (one seed). This is expected, because OARF gives up in-distribution accuracy for interventional robustness and in-distribution loss does not reward that. The Diebold–Mariano statistics on the synthetic benchmark are large (Figure 7).

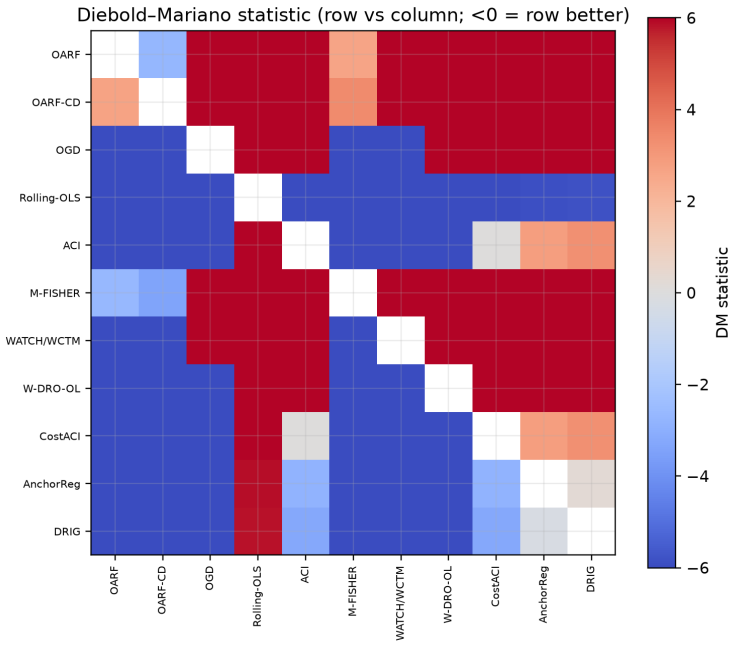


Figure 7: Pairwise Diebold–Mariano statistics on the synthetic benchmark; a negative entry means the row method is significantly more accurate than the column method.

5 Weekly energy volatility

The simulation tells us what OARF can do when its assumptions hold. This section asks whether any of that carries over to a real forecasting problem in which the assumptions can at best hold approximately.

Data and target. We use a public daily panel (Table 8) aligned to business days with a forward-fill capped at five days. The targets are the price volatility of Brent crude and of Henry Hub natural gas. Only daily closing prices are freely available, so intraday realised variance cannot be formed. We therefore forecast weekly realised variance from daily returns: for the week ending on Friday w, the target is $y_w = \log \sum_{s \in w} r_s^2$, the log of the sum of squared daily log returns in that week, and every forecast of y_w uses data up to the end of week $w-1$ only. Consecutive weeks do not overlap, so no part of the target is known when the forecast is made. This choice matters. A daily target built from a trailing five-day mean of squared returns shares four of its five terms with its own lag; on Brent its correlation with yesterday's value is 0.89, against 0.53 for the weekly target, and most of its apparent predictability is mechanical. Appendix B reports the daily design for comparison. The sample runs from April 2010 to the week of 26 June 2026 (847 weeks); the first 169 weeks are the warm-up and the remaining 678 (from July 2013) are evaluated.

The anchor and what it assumes. The anchor is the end-of-week level of three uncertainty indices: the CBOE VIX, the daily Geopolitical Risk index and the daily Economic Policy Uncertainty index. The other drivers (the dollar exchange rate, the ten-year yield, their weekly changes and lagged volatility) enter the context but not the anchor. For OARF to help, the regime shifts that matter for energy volatility would have to move the level of these indices, and the indices would have to behave roughly like the exogenous anchor of Proposition 1. The two news-based

Table 8: The public real-data panel. All series are free and are fetched by the released download script; the version used here was downloaded on 4 November 2026.

Series	Source / ticker	Span	Role
Brent crude	FRED `DCOILBRENTEU`	2010–2026	target
Henry Hub gas	FRED `DHHNGSP`	2010–2026	target
VIX	FRED `VIXCLS`	2010–2026	driver, anchor
EUR/USD	FRED `DEXUSEU`	2010–2026	driver
10-year Treasury yield	FRED `DGS10`	2010–2026	driver
Geopolitical Risk index	Caldara and Iacoviello (2022)	2010–2026	driver, anchor
Economic Policy Uncertainty	Baker et al. (2016)	2010–2026	driver, anchor

indices are constructed from newspaper text and are not computed from energy prices, which makes them the more plausible candidates. The VIX is a market price that responds to many of the same shocks as commodity volatility, so it is the least plausible. None of the three is known to be the channel through which energy-volatility regimes travel. The stress tests of Section 4 tell us what to expect if the anchor is only weakly aligned with the true channel or observed with noise: little or no gain, and possibly a small loss. The real study is therefore a test of whether these indices are good enough anchors, as well as a test of the method.

Forecast pool and benchmarks. Every method combines the same seven one-week-ahead forecasts. They are last week's log realised variance; its means over the previous 4 and 13 weeks, in the spirit of the HAR model (Corsi, 2009); a RiskMetrics exponentially weighted variance and a GARCH$(1,1)$ variance, both computed from daily returns up to the end of the previous week and scaled to five days; the log VIX at the end of the previous week; and a rolling ridge regression on five lags of the target and the lagged drivers, whose features are standardised with the statistics of its own training window. The methods are OARF, OARF-CD and nine of the baselines of Table 3 (WATCH/WCTM, CostACI and DRIG are not run on the real data), with the hyper-parameters of Table 4; nothing was re-tuned for the weekly data. We add two fixed benchmarks that a forecasting reader would expect: persistence, which forecasts this week's log variance by last week's, and the equal-weight average of the six members of the pool that forecast log variance directly (the log VIX is on a different scale and is left out). Both benchmarks are scored under the same protocol as the learning methods. Regimes for the regime-aware metrics are terciles of the lagged four-week mean of the target, with segments shorter than eight weeks merged into their neighbours. The tercile thresholds use the full sample; the labels are used only to score forecasts and no method sees them.

Results. Table 9 reports the results, and Figure 8 shows the Brent error path. The first finding is that the combination methods are useful. On both series every learned combination has a lower MSE than persistence; for most methods the reduction is about a third on Brent and a fifth on Henry Hub. On Brent the Model Confidence Set excludes persistence ($p < 0.01$) and the expected-shortfall combiner ($p = 0.06$) and retains the other eleven methods. On Henry Hub it retains all thirteen, persistence included ($p = 0.17$), so the gain over persistence there is not significant at the 10% level.

The second finding is that OARF adds nothing on these data. Its MSE is within 1.5% of online gradient descent on both series (Brent 0.714 against 0.704, Henry Hub 1.014 against 1.009), it is slightly worse in the worst regime, and the equal-weight average does as well as either. The batch anchor-regression reference does best on Henry Hub (0.984) and among the worst on Brent

(0.741). None of these differences is significant. Apart from persistence on Henry Hub (0.17) and anchor regression on Brent (0.52), every retained method has an MCS p-value of 1.00. The pairwise Diebold–Mariano test of OARF against online gradient descent gives $p = 0.47$ on Brent and $p = 0.54$ on Henry Hub (Figure 9). Some pairs of baselines do differ in the pairwise test, for example ACI and W-DRO-OL on Brent, but the Model Confidence Set, which accounts for the number of comparisons, keeps them together. The coverage of the conformal head is 0.86 on Brent and 0.87 on Henry Hub, against a nominal 0.90.

This is the outcome the stress tests predict if the uncertainty indices are noisy or weakly aligned proxies for whatever moves energy volatility, and the data do not let us separate that explanation from the alternative that energy-volatility regimes are not anchor-mediated mean shifts at all. Two secondary checks give the same picture. A volatility-timing strategy built on each forecast earns low Sharpe ratios whose ranking does not follow forecast accuracy (Appendix C). In two event windows, the COVID-19 crash of 2020 and the energy shock of 2022, OARF has a lower cumulative loss than online gradient descent in three of the four window–commodity pairs and a higher one in the fourth, the 2022 window for Brent (Appendix F).

Table 9: Weekly log realised variance, non-overlapping weeks, evaluation over the last 678 of 847 weeks (2013–2026). MSE is on the target's past-only standardised scale, so persistence and the learned combinations are directly comparable; w-reg is the worst-regime MSE and post-cp the MSE in the 50 weeks after each regime change. MCS p is the Model Confidence Set p-value (Hansen et al., 2011); methods with $p < 0.10$ are outside the 10% set. The two fixed benchmarks are listed first.

	Brent				Henry Hub			
Method	MSE	w-reg	post-cp	MCS p	MSE	w-reg	post-cp	MCS p
Persistence	1.072	1.306	1.025	< 0.01	1.267	1.664	1.448	0.17
Equal-weight average	0.708	0.802	0.687	1.00	1.013	1.269	1.107	1.00
OARF	0.714	0.823	0.689	1.00	1.014	1.287	1.124	1.00
OARF-CD (Section 6)	0.713	0.800	0.710	1.00	0.987	1.264	1.108	1.00
OGD	0.704	0.804	0.687	1.00	1.009	1.261	1.129	1.00
Rolling-OLS	0.712	0.826	0.685	1.00	1.047	1.301	1.208	1.00
ACI	0.720	0.831	0.697	1.00	0.995	1.256	1.106	1.00
M-FISHER (2025)	0.730	0.856	0.715	1.00	1.110	1.311	1.227	1.00
DR-Combo (Wang 2026)	0.723	0.815	0.714	1.00	1.054	1.302	1.184	1.00
FC-DRO (Liu 2026)	0.713	0.800	0.696	1.00	1.088	1.276	1.177	1.00
FC-DRO-ES (Liu 2026)	0.898	0.985	0.847	0.06	1.134	1.503	1.299	1.00
W-DRO-OL (Chen 2026)	0.704	0.804	0.687	1.00	1.010	1.263	1.129	1.00
AnchorReg (rolling batch)	0.741	0.844	0.706	0.52	0.984	1.230	1.125	1.00

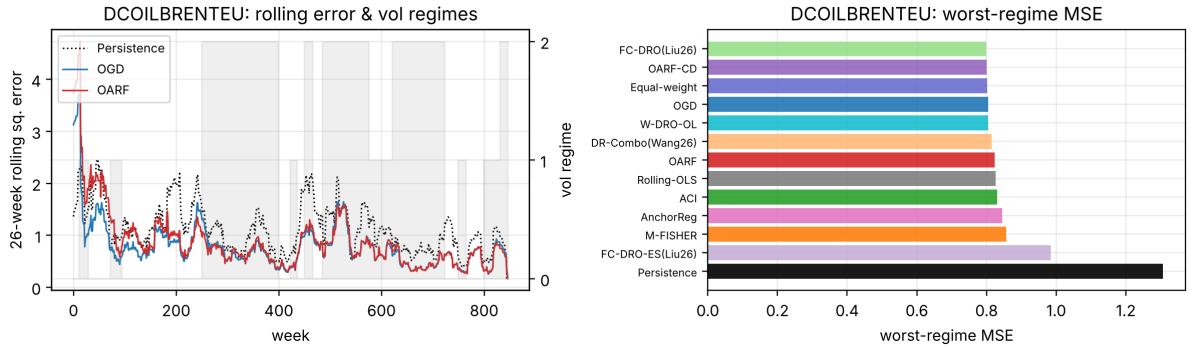


Figure 8: Brent, weekly design. Left: 26-week rolling squared error of persistence, OGD and OARF, with the volatility regime shaded. Right: worst-regime MSE by method.

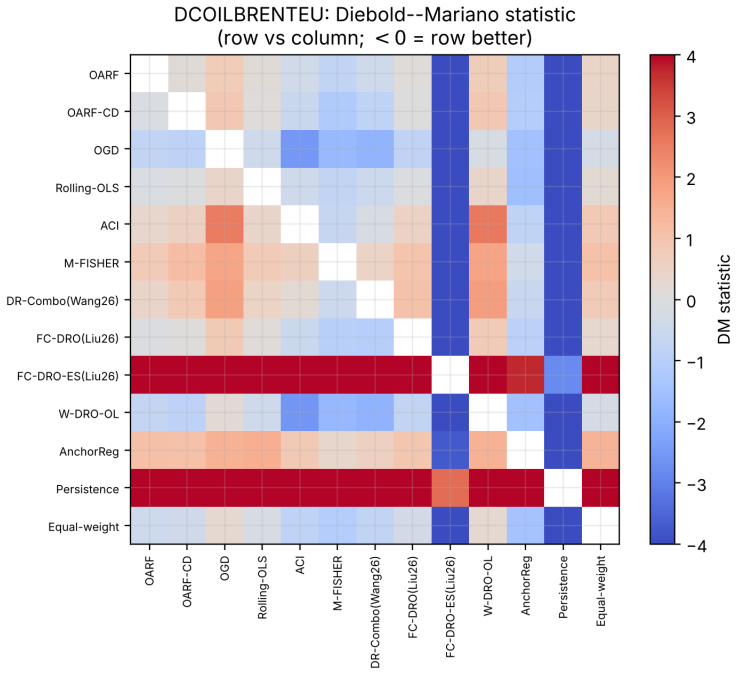


Figure 9: Brent, weekly design: pairwise Diebold–Mariano statistics (negative means the row method is more accurate than the column method). Persistence and the expected-shortfall combiner are clearly worse than the rest; among the other methods most statistics are small.

6 Extension: learning the anchor

So far the analyst has supplied the anchor. In practice the channel is often uncertain: Section 5 used three candidate indices without knowing whether any of them carries the regime. This section describes a rule, OARF-CD, that estimates the channel from the stream, a statistic that indicates when the estimate can be trusted, and the evidence on both. Theorem 1 does not cover the learned channel.

6.1 Method

Online channel discovery. The rule estimates B from the stream. It performs unsupervised nonstationary-subspace estimation, not causal discovery. Under the SCM of Proposition 1, a regime acts on the context by shifting the anchor's location, so the anchor directions are nonstationary in mean across regimes and stationary nuisance directions are not. We therefore track the across-regime dispersion of the context's conditional mean and ascend it on a slow timescale $\eta_B \ll \eta$,

$$\max_{\|B\|_F \le 1} \mathcal{D}(B) = \operatorname*{Var}_{\text{regimes}} \big(\mathbb{E}[B^\top z \mid \text{regime}]\big) - \gamma_c \left\|\mathbb{E}[B^\top z]\right\|^2. \tag{6}$$

This objective selects high-dispersion nonstationary directions. It does not by itself establish that they are exogenous or causal. The grand-mean penalty (second term) discourages loading on the persistent component but does not guarantee exogeneity, and a direction that is only volatile or trending can be selected; the stress tests of Section 4 probe this failure mode. What the layer recovers is the subspace through which regimes appear to act in the observable context, and this coincides with the causal channel only under the SCM assumptions.

Streaming (6) needs a proxy for "regime". We use a forgetting partition into length-L blocks. A block should be long enough for a stationary driver's short-horizon autocorrelation to average out within it (so L should exceed the dominant driver autocorrelation time), and short enough that actual regime changes fall across block boundaries. The block length L is a hyperparameter selected on the warm-up window. Let S be the across-block scatter of the block-mean context and u its grand mean. The ascent direction is $\nabla_B \mathcal{D} = 2(S - \gamma_c uu^\top)B$. At each block boundary B is updated after that step's robust gradient step and re-orthonormalised onto the Frobenius ball, and the new B is used from the next step onward. A lightweight non-robust reference predictor is run alongside to keep discovery oriented toward predictively relevant directions. Algorithm 1 gives the complete loop, with discovery as an optional inner block.

Reliability diagnostic. The discovery layer can fail silently. Nothing in Algorithm 1 tells the forecaster whether the current B is a good estimate of the true channel, and in deployment the true channel is unobserved. As a proxy for how well the top-q subspace is identified, we use a quantity the layer already computes. Let $\lambda_1 \ge \cdots \ge \lambda_p$ be the eigenvalues of $M = S - \gamma_c uu^\top$ at a given block boundary (Eq. (6)), and define the *normalised eigengap*

$$\kappa \;=\; \frac{\lambda_q - \lambda_{q+1}}{\sum_{i=1}^{p} |\lambda_i| + \epsilon} \in [0, 1]. \tag{7}$$

By the Davis–Kahan $\sin\Theta$ theorem, the angle between the top-q eigenspace of an estimated symmetric matrix and that of the population matrix it estimates is controlled by the corresponding population eigengap. A small κ means that the q-dimensional subspace B is ascending is only weakly separated from the rest of the spectrum, so which q directions are selected depends on sampling noise. A large κ means the top-q subspace is well separated and stably identified. The

subspace alignment used below to score the discovery layer against ground truth needs the true channel, whereas κ is computed entirely from the stream and is available in a live deployment. Davis–Kahan bounds estimation error given a population gap; it says nothing about whether the recovered subspace is exogenous or causal, which still rests on the SCM assumptions of Proposition 1. The evidence below tests whether κ tracks true recovery and whether gating on it helps.

6.2 Evidence

Learned channel. When it has to find its own anchor, OARF-CD recovers the true subspace on average. Mean principal-subspace alignment is 0.88 ± 0.16, against a random-subspace baseline of 0.42, and individual seeds climb toward 0.99 as the stream lengthens (Figure 10). Its median worst-do(A) MSE (0.049) is close to that of fixed-channel OARF (0.041, Table 5), but the distribution has a heavy tail. The mean across seeds is 0.147, roughly three times the median, and on a minority of seeds the error approaches or exceeds that of OGD (median 0.095), which applies no anchor penalty. A seed's final alignment is only noisily related to its worst-do(A) MSE (Figure 11), and the seeds with poor alignment account for the tail. The layer therefore recovers a nonstationary subspace but does not match the robustness of a known channel, and this tail is its main practical limitation.

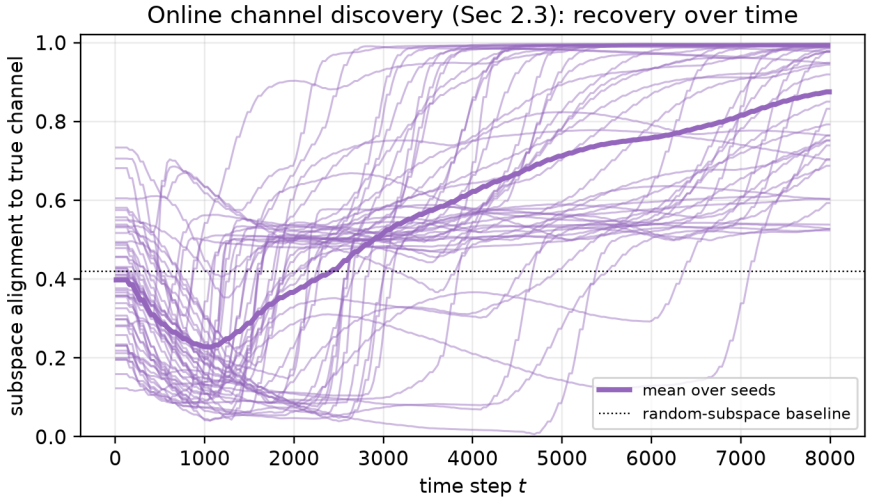


Figure 10: Online channel discovery: alignment of the learned channel B_t to the true intervention subspace over time. Thin lines are individual seeds, the bold line is their mean, and the dotted line is the random-subspace baseline.

Online reliability diagnostic. Figure 11 shows that the failures of the discovery layer are a discrete tail risk. Most seeds recover a well-aligned channel, a minority do not, and those seeds dominate the mean worst-case error. If they could be flagged while the stream runs, without consulting the true channel (which a deployment never observes), the layer could decline to trust its own channel on those streams. Figure 12a tests whether the eigengap κ of Eq. (7) can serve as such a flag. Across the 50 seeds, κ correlates with the true subspace alignment, which is otherwise unobservable, at $r = 0.62$, and 13 of the 16 seeds whose final alignment falls below 0.8 also fall in the bottom quartile of post-warm-up eigengap. Figure 12b uses this as a gate. When a seed's average post-warm-up κ lies in the bottom quartile (only 13 of 50 seeds), the forecaster reverts

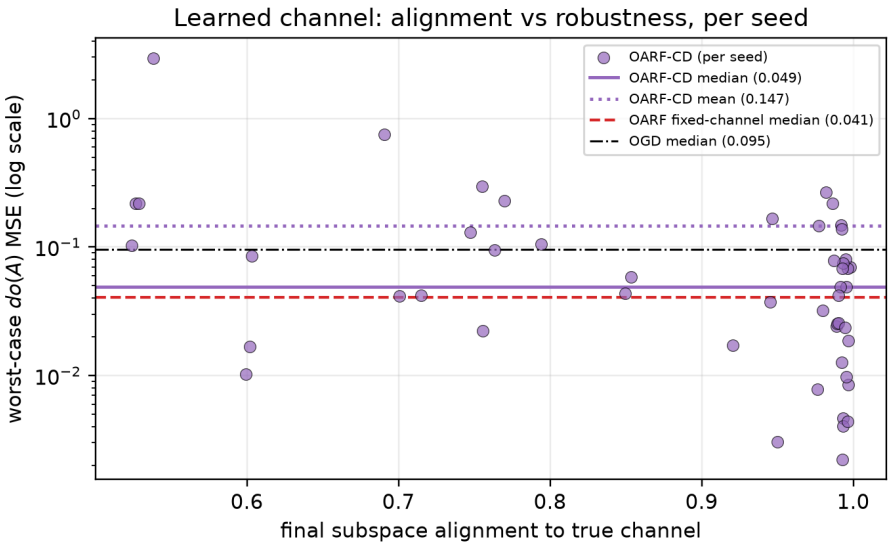


Figure 11: Per-seed alignment versus interventional robustness for the learned channel (50 seeds). Each point is one seed: the horizontal axis is the final subspace alignment of OARF-CD's discovered channel and the vertical axis its worst-case do(A) MSE (log scale). High-alignment seeds cluster at low error near the fixed-channel OARF median (dashed), but a minority of low-alignment seeds incur an order-of-magnitude higher error. These seeds pull the mean (dotted) far above the median (solid) and make the learned channel unreliable in the worst case.

to plain OGD; otherwise it keeps the learned channel. Gating lowers the mean worst-do(A) MSE of OARF-CD from 0.147 (ungated) to 0.113, a reduction of about a quarter, and closes about two-fifths of the gap to the 0.059 attained with the true channel. The gate uses only a statistic the discovery layer already computes. Its threshold, however, is the 25th percentile of κ across these same 50 seeds, so in deployment it would have to be calibrated in advance, for example on simulated streams. The decision uses the average of κ over the second half of each stream, so the gate makes one choice per stream. It also discards a misspecified channel instead of correcting it; like Davis–Kahan, a large gap certifies a well-identified subspace but says nothing about whether that subspace is exogenous.

Comparison of anchor selectors. Automatic anchor selection is the most fragile part of the method, so we compare the block-mean rule of OARF-CD with alternative selectors under a strict pre-deployment budget: the channel is estimated on the warm-up block only, with no look-ahead (Table 10). The alternatives are a supervised residual–context-coupling selector, stationary PCA of the context, and a random channel. Each is scored by its alignment to the truth and by the worst-do(A) MSE of the resulting fixed-channel method. The block-mean rule does about as well as the supervised selector and much better than stationary PCA. From a short warm-up, though, neither is clearly better than a random channel: their worst-do(A) MSEs are 0.101 and 0.093 against 0.115 for a random channel and 0.070 for the true one, and their alignments (0.42 and 0.37) are no higher than the random channel's (0.43). High alignment is reached only by learning over the full stream (OARF-CD, alignment 0.84 in Table 6), and even then the estimate is unstable. Reliable online anchor discovery from limited data thus remains an open problem. Practitioners should supply a candidate anchor and validate it with the procedure of Section 7 instead of relying on automatic

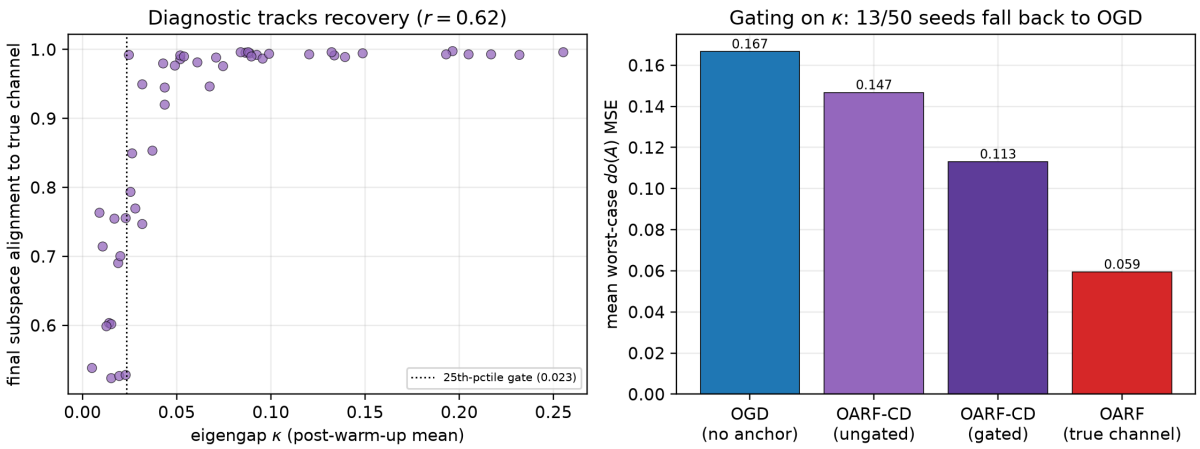


Figure 12: Eigengap diagnostic. (a) Per-seed post-warm-up eigengap κ vs. true subspace alignment ($r = 0.62$); the dotted line is the 25th-percentile gate. (b) Mean worst-case do(A) MSE for OGD, ungated OARF-CD, OARF-CD gated on κ, and fixed-channel OARF with the true channel. Gating reduces the ungated learned channel's mean worst-case error by about a quarter without using the ground truth.

discovery alone.

Table 10: Anchor selectors estimated on the warm-up block only (8 seeds; alignment to the true channel, and worst-do(A) MSE of the resulting fixed-channel OARF; lower MSE / higher alignment better).

Selector	subspace alignment	worst-do(A) MSE (median)
true channel (reference)	1.00	0.070
block-mean (OARF-CD rule)	0.42	0.101
supervised residual–context coupling	0.37	0.093
stationary PCA of context	0.31	0.236
random channel	0.43	0.115

Real data. On the energy data the learned channel does about as well as the hand-chosen one (Table 9: MSE 0.713 against 0.714 on Brent and 0.987 against 1.014 on Henry Hub), and both are inside the Model Confidence Set. Since neither improves on online gradient descent, these numbers say nothing about whether the learned channel is closer to the true one.

7 Discussion and conclusion

When a regime shift moves an observed anchor and the forecasting relation is confounded through that anchor, a forecaster can be made robust to the shift in advance and online. In our simulation the gain is large: OARF more than halves the worst-case error of online gradient descent, and it pays for this with a higher in-distribution error that the penalty ξ controls. Distributionally robust methods spread their protection over all directions, while anchoring concentrates it on the directions in which the anchor moves. This helps only if those directions are the right ones. With

an anchor observed with noise of about one standard deviation, or misaligned with the true channel by more than about 50°, the benefit is gone, and when regimes shift variances or the target equation the method does worse than no anchor.

The energy application shows how far this carries to real data. With a target that does not overlap its own lag, the learned combinations beat persistence, but OARF with macro-uncertainty indices as the anchor is no better than online gradient descent or an equal-weight average. We read this as evidence that these indices are, at best, noisy and weakly aligned proxies for whatever moves energy volatility, which is the case in which our stress tests predict no gain; the data cannot rule out that energy regimes are simply not anchor-mediated mean shifts. The earlier daily design, whose target overlapped its lag, made all methods look more accurate than they are and could not separate them from persistence (Appendix B).

Using the method. These results suggest a simple procedure. Propose candidate anchors from domain knowledge. On the warm-up block, check that the correlation between the anchor and the residual of a plain online learner is non-trivial and stable, and run a stress test that perturbs the anchor as in Figure 6. Use fixed-channel OARF only if anchoring improves the validation worst-regime loss at an acceptable in-distribution cost and a random or noisy version of the anchor does not do as well. Otherwise use plain online learning. Treat a learned channel as a diagnostic unless its eigengap is large, and do not use the method when the regime shifts appear as changes in variance or in the target equation.

Limitations. The theorem covers the fixed channel with an Ader step wrapper, while the experiments use AdaGrad (Appendix E), and its $T^{2/3}$ rate for estimated moments relies on stationarity within regimes. Tightening this rate toward $\sqrt{T}$ without a changepoint trigger, and extending a guarantee to a learned or misspecified channel, remain open; the Davis–Kahan bound behind the eigengap check supplies only the estimation half of such an argument. The real study has 678 evaluation weeks, uses a variance proxy from daily closing prices because intraday data are not freely available, and uses an equity-market implied volatility in the absence of a free commodity-specific one. Other extensions are nonlinear anchors and predictors and richer interventions on variances and mechanisms.

Reproducibility. All code and data-build scripts that regenerate every number, table and figure in this paper are available at `https://github.com/vinhqdang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift`.

A Proof of Theorem 1

Throughout, $\mathcal{R}_t(w) := \sup_{\|\nu\| \le \rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}[(y_t - w^\top \phi(x_t))^2]$ is the per-step worst-case risk in (5), and $\hat{g}_t$ is the (inexact) step direction of Algorithm 1.

Lemma 1 (Convex surrogate and gradient decomposition)**.** *$\mathcal{R}_t$ is convex, and under Assumption 2 it equals, within the active regime, the anchor-penalised risk $\mathcal{R}_t(w) = \mathbb{E}_t[(y - w^\top \phi)^2] + \xi P_t(w)$ of (1). Also, $\hat{g}_t = \nabla \mathcal{R}_t(w_t) + \zeta_t$ with $\zeta_t = M_t + b_t$, where $M_t := -r_t \phi(x_t) - \mathbb{E}_t[-r\phi]$ is a martingale difference ($\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$) and b_t is the deterministic ($\mathcal{F}_{t-1}$-measurable) error of the EMA-estimated penalty gradient relative to $\nabla(\xi P_t)$.*

Proof. For fixed ν, $\mathbb{E}_{\mathrm{do}(a:=\nu)}[(y - w^\top \phi)^2]$ is a convex quadratic in w; a pointwise supremum of convex functions is convex. The identity is Proposition 1 within the stationary active regime. The

prediction term $-r_t\phi(x_t)$ is an unbiased one-sample estimate of $\frac{1}{2}\nabla\mathbb{E}_t[(y - w^\top\phi)^2] = -\mathbb{E}_t[r\phi]$, with centred part M_t; the robustness term substitutes the $\mathcal{F}_{t-1}$-measurable EMA moments for the population moments in $\nabla(\xi P_t)$, and b_t is that substitution error. □

Lemma 2 (Inexact dynamic regret)**.** *Run projected OGD with the path-length-adaptive Ader wrapper (Zhang et al., 2018) on the convex losses $\mathcal{R}_t$, fed $\hat{g}_t$. Then for any $u_{1:T} \subset \mathcal{W}$ of path length P_T, $\sum_t \mathcal{R}_t(w_t) - \sum_t \mathcal{R}_t(u_t) \le O(\sqrt{T(1+P_T)}) + \sum_t\langle\zeta_t, w_t - u_t\rangle$, with the first term attained without prior knowledge of P_T.*

Proof. With exact gradients, Ader attains minimax dynamic regret $O(\sqrt{T(1+P_T)})$ for convex G-Lipschitz losses over a radius-D set, adaptively in P_T (Zhang et al., 2018), subsuming Zinkevich (2003); Hazan (2016). Convexity gives $\mathcal{R}_t(w_t) - \mathcal{R}_t(u_t) \le \langle\nabla\mathcal{R}_t(w_t), w_t - u_t\rangle = \langle\hat{g}_t, w_t - u_t\rangle - \langle\zeta_t, w_t - u_t\rangle$; summing and applying the exact-gradient bound to the $\hat{g}_t$ terms yields the inexactness term. □

Lemma 3 (Gradient-error control)**.** *Under Assumptions 1–4, against the deterministic minimising comparator: (i) $\mathbb{E}[\sum_t\langle M_t, w_t - u_t\rangle] = 0$; and (ii) $\mathbb{E}\|b_t\| \le c_1\sqrt{1-\beta}$ at steps at least $1/(1-\beta)$ into a regime, with $\|b_t\| \le c_2$ within $1/(1-\beta)$ steps of each of the K changes, so $\sum_t \mathbb{E}\|b_t\| \le c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta)$; with oracle moments $b_t \equiv 0$.*

Proof. (i) w_t is $\mathcal{F}_{t-1}$-measurable and, the schedule being oblivious (Assumption 3), u_t is deterministic, so $\mathbb{E}[\langle M_t, w_t - u_t\rangle \mid \mathcal{F}_{t-1}] = 0$; the tower rule concludes. (ii) An EMA of decay β of a stationary sequence has variance $\Theta(1-\beta)$ about its mean and geometrically-decaying within-regime bias; by Assumption 1 the penalty gradient is Lipschitz in these moments (the ϵ-ridge bounds the inverse), inheriting the $\Theta(\sqrt{1-\beta})$ steady-state scale. After a change the EMA mixes the previous regime for $O(1/(1-\beta))$ steps with bounded $\|b_t\| \le c_2$. Summing gives the bound; oracle moments make the substitution error vanish. □

Proof of Theorem 1. Combining the lemmas and taking expectations, for the deterministic minimiser $u_{1:T} \in \mathcal{W}$ ($P_T \le 2DK$), $\mathbb{E}[\sum_t \mathcal{R}_t(w_t)] - \sum_t \mathcal{R}_t(u_t) \le O(\sqrt{T(1+K)}) + 0 + D\sum_t \mathbb{E}\|b_t\|$. With oracle moments the last term vanishes, giving (i); with EMA moments it is $\le D(c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta))$, which at $1-\beta_T = T^{-2/3}$ is $O((1+K)T^{2/3})$ and dominates $\sqrt{T(1+K)}$, giving (ii). Finally, by Assumption 2 the realised regime is an admissible intervention, so $\mathbb{E}_t[\ell_t(w_t) \mid \mathcal{F}_{t-1}] \le \mathcal{R}_t(w_t)$ and hence $\mathbb{E}[\sum_t \ell_t(w_t)] \le \mathbb{E}[\sum_t \mathcal{R}_t(w_t)]$; as $\min_{u\in\mathcal{W}}\sum_t \mathcal{R}_t(u)$ is the comparator in (5), the displays bound $\mathbb{E}[\mathrm{IntReg}_T(\rho(\xi))]$ by the stated rates. □

B The daily design with an overlapping target

An earlier version of the real study forecast, one trading day ahead, the log of a trailing five-day mean of squared daily returns, $\log\left(\frac{1}{5}\sum_{k=0}^{4} r_{t-k}^2\right)$. Four of the five terms are known when the forecast is made, so the target is close to its own lag (correlation 0.89 on Brent) and much of its predictability is mechanical. Table 11 repeats that design with the two benchmarks of Section 5. The MSEs are much lower than in Table 9 because of the overlap. Persistence, which repeats yesterday's value of the target, is inside the Model Confidence Set with $p = 1.00$ on both series, so in this design no method, OARF included, is significantly more accurate than the simplest forecast. The equal-weight average does badly here, probably because it mixes the overlapping one-day lag with monthly averages. This comparison is the reason for the weekly design of the main text.

Table 11: The earlier daily design, whose target is the log of a trailing five-day mean of squared returns, so that four of its five terms are known one step ahead. Same methods and protocol as Table 9, evaluation over the last 3,408 of 4,260 trading days. Persistence (yesterday's value of the target) is inside the Model Confidence Set with $p = 1.00$ on both series.

	Brent				Henry Hub			
Method	MSE	w-reg	post-cp	MCS p	MSE	w-reg	post-cp	MCS p
Persistence	0.243	0.253	0.271	1.00	0.290	0.389	0.244	1.00
Equal-weight average	0.309	0.311	0.312	< 0.01	0.452	0.546	0.426	< 0.01
OARF	0.228	0.242	0.249	1.00	0.287	0.391	0.263	1.00
OARF-CD (Section 6)	0.227	0.240	0.251	1.00	0.290	0.388	0.269	1.00
OGD	0.225	0.239	0.246	1.00	0.286	0.387	0.267	1.00
Rolling-OLS	0.221	0.232	0.242	1.00	0.285	0.387	0.260	1.00
ACI	0.218	0.229	0.244	1.00	0.278	0.379	0.242	1.00
M-FISHER (2025)	0.275	0.279	0.305	0.01	0.354	0.446	0.310	0.01
DR-Combo (Wang 2026)	0.230	0.239	0.254	1.00	0.286	0.391	0.241	1.00
FC-DRO (Liu 2026)	0.232	0.241	0.258	1.00	0.285	0.387	0.241	1.00
FC-DRO-ES (Liu 2026)	0.249	0.316	0.303	0.13	0.307	0.407	0.274	0.24
W-DRO-OL (Chen 2026)	0.225	0.239	0.246	1.00	0.285	0.387	0.264	1.00
AnchorReg (rolling batch)	0.218	0.226	0.241	1.00	0.288	0.389	0.270	1.00

C Decision-economic evaluation

We convert each method's weekly variance forecast into a volatility-timing portfolio (Fleming et al., 2001). Exposure is scaled inversely to the forecast variance, $w_t = \mathrm{clip}(\sigma^{\star 2}/\hat{\sigma}_t^2, 0, 3)$, with a weekly volatility target $\sigma^\star = 0.01\sqrt{5}$, and the strategy earns $w_t r_{t+1}$, the next week's return, net of a one-basis-point turnover cost. Forecasts are mapped back to the variance scale with the target standardiser in force at each week, so the strategy uses no future information. Over the evaluation period the annualised Sharpe ratios are negative for every method on Brent (between -0.48 and -0.07; persistence has the highest) and between 0.08 and 0.30 on Henry Hub (Rolling-OLS highest, persistence lowest). The ranking does not follow forecast accuracy, and we do not test the differences, so this evaluation is reported for completeness only (Figure 13).

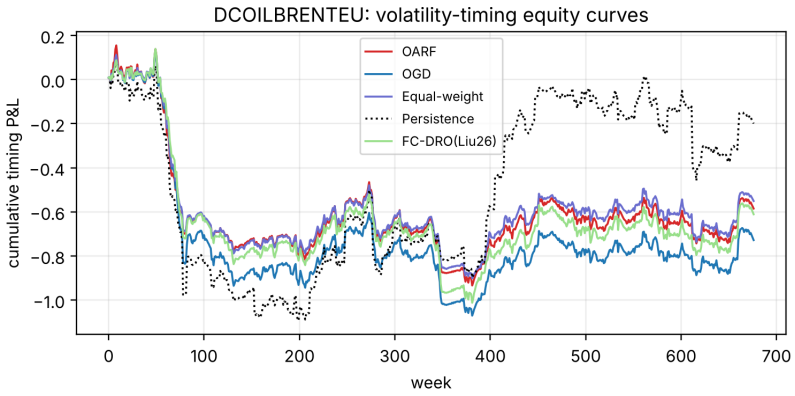


Figure 13: Brent, weekly design: cumulative volatility-timing profit and loss for selected methods.

D Baseline implementations and notes

Table 12 lists, for each method, the loss optimised, the update rule and the output, whether the method runs online, rolling or in batch, and whether it uses the anchor. All methods run under the same protocol with the hyper-parameters of Table 4. Per step, the linear methods cost $O(d)$ and OARF adds $O(pq + q^3)$ for the anchor block. Rolling-OLS and the batch methods cost $O(d^3)$ at each refit, which takes place every 5–10 steps.

Some implementation details follow. The warm-up block is used only to initialise the past-only standardiser and to select hyper-parameters once; it is never used for separate batch pre-training. Missing FRED observations and exchange holidays are handled by reindexing to business days with a forward-fill capped at five days, and longer gaps are left missing and skipped. For the weekly design the daily data are aggregated to weeks ending on Friday, and weeks with fewer than two trading days are dropped. The GARCH$(1, 1)$ expert is a recursion with fixed coefficients ($\alpha = 0.08$, $\beta = 0.90$) and is never estimated, so it uses no future data. The channel dimension is larger on the real data ($q = 3$, against $q = 2$ in the synthetic experiments) because three macro-uncertainty drivers (VIX, GPR, EPU) are chosen by hand as the anchor, while the synthetic SCM has exactly two anchor directions by construction. We present OARF for linear predictors. It extends to nonlinear forecasters by applying the same residual–anchor penalty to a feature map or to the top layer of a network; we leave this extension to future work.

Table 12: Baseline implementations. "Out" = P(oint)/I(nterval); "Mode" = O(nline)/R(olling)/B(atch); "Anc" = uses the anchor.

Method	Loss optimised	Update rule	Out	Mode	Anc
OARF/OARF-CD	anchor-penalised sq. err.	AdaGrad on (4)	P	O	yes
OGD	squared error	AdaGrad ($\xi = 0$)	P	O	no
Rolling-OLS	windowed ridge	closed-form refit	P	R	no
ACI/CostACI	pinball (3 quantiles)	subgrad. + ACI level	P+I	O	no
M-FISHER	sq. err. + martingale	OGD + triggered NGD	P	O	no
WATCH/WCTM	sq. err. + WCTM	OGD + width adapt	P+I	O	no
DR-Combo	Wasserstein-DR combo	EG mirror descent	P	O	no
FC-DRO(-ES)	variance-/ES-scaled	exponential weights	P	O	no
W-DRO-OL	Wasserstein-DR sq. err.	primal–dual OGD	P	O	no
AnchorReg	anchor-penalised (batch)	windowed closed-form	P	R	yes
DRIG	invariant-gradient (batch)	windowed two-env. fit	P	R	(env)

E Ader-wrapped versus AdaGrad OARF

The interventional-regret theorem (Theorem 1) is stated for the path-length-adaptive Ader step wrapper (Zhang et al., 2018), whereas the deployed model takes a single AdaGrad step. To check whether this simplification changes the empirical behaviour, we implemented the Ader variant and compared it with the deployed AdaGrad model on the controlled anchor-SCM (true fixed channel, $\xi = 6$, 12 seeds). The Ader variant is a pool of five gradient-descent experts that follow the same anchor-robust surrogate gradient with a geometric ladder of step sizes. Each expert is projected onto the comparator ball, and a Hedge meta-learner combines them. Table 13 reports the results. Both step rules cut the worst-case do(A) error well below the no-anchor OGD reference, to about a third (AdaGrad) and a half (Ader) of it, and their per-seed worst-case errors are strongly rank-correlated (Spearman $\rho = 0.80$). In this experiment AdaGrad is slightly more accurate and has lower variance. It also has fewer parameters, and for these reasons it is the rule we deploy; the Ader wrapper is the object the theorem analyses. Substituting one for the other leaves the qualitative ranking unchanged.

Table 13: Ader-wrapped versus AdaGrad OARF on the synthetic anchor-SCM (12 seeds, true fixed channel, $\xi = 6$; median with $[\, q_{25}, q_{75} \,]$). Both reach similar worst-case interventional error, far below the OGD reference, and their per-seed worst-case errors are rank-correlated at Spearman $\rho = 0.80$.

Step rule	in-regime MSE	worst-do(A) MSE
AdaGrad (deployed)	0.028	$0.052\,[0.030, 0.114]$
Ader (analysed)	0.030	$0.073\,[0.026, 0.142]$
OGD ($\xi = 0$, reference)	0.019	$0.143\,[0.065, 0.353]$

F Event study: COVID-19 and the 2022 energy shock

The regime-tercile analysis of Section 5 is aggregate, so here we examine two dated episodes on the weekly design: January to September 2020 (39 weeks) and calendar 2022 (52 weeks). We run fixed-anchor OARF ($\xi = 1.5$) and the no-anchor OGD reference, and within each window we plot the

cumulative squared-loss gap (OARF minus OGD, so a negative value means OARF is ahead) and, for both methods, the smoothed anchor–residual correlation $|\mathrm{corr}(a_t, r_t)|$ (Figure 14). OARF is ahead in three of the four window–commodity pairs: the COVID-19 window on Brent (cumulative gap -0.77) and both windows on Henry Hub (-1.16 and -2.46). It is behind in the 2022 window on Brent ($+2.11$). In the three windows where it leads, OARF keeps the anchor–residual correlation below that of OGD in most weeks (67%, 79% and 100% of weeks); in the 2022 Brent window it does so in only 4% of weeks. With two episodes and no significant aggregate difference, this is an illustration and not evidence of an advantage.

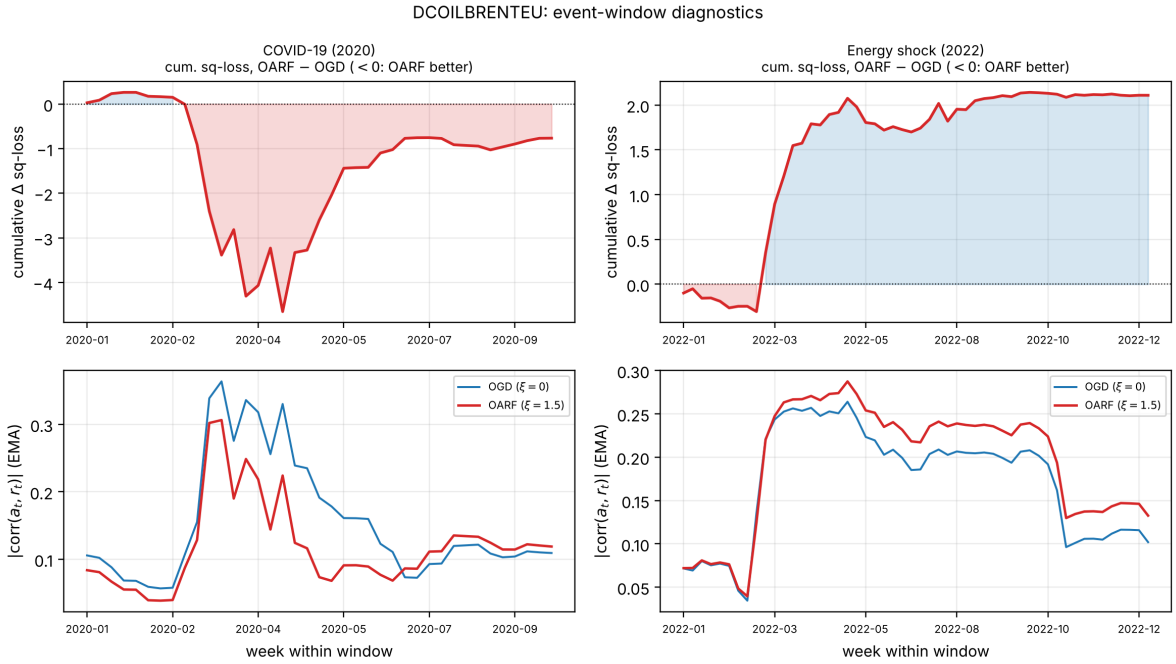


Figure 14: Brent event windows, weekly design. Top: cumulative squared-loss gap, OARF minus OGD (shaded where OARF leads or trails). Bottom: the smoothed anchor–residual correlation of both methods.

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Conflict of interest statement

The author declares no conflict of interest.

Data availability statement

The data that support the findings of this study are openly available from the public sources listed in Table 8. The code and the scripts that download these data and regenerate every number, table and figure are openly available at `https://github.com/vinhqdang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift`.

References

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131(4):1593–1636.

Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225.

Chen, G., Fattahi, S., and Shafiee, S. (2026). Wasserstein distributionally robust online learning. *arXiv preprint arXiv:2602.20403*.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Duchi, J., Hazan, E., and Singer, Y. (2011). Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12:2121–2159.

Fleming, J., Kirby, C., and Ostdiek, B. (2001). The economic value of volatility timing. *The Journal of Finance*, 56(1):329–352.

Gibbs, I. and Candès, E. (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 1660–1672.

Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2):453–497.

Hazan, E. (2016). Introduction to online convex optimization. *Foundations and Trends in Optimization*, 2(3–4):157–325.

Khan, B. and Syed, T. Q. (2025). Technical note on sequential test-time adaptation via martingale-driven Fisher prompting. *arXiv preprint arXiv:2510.03839*.

Liu, J. and Cheng, M.-Y. (2026). Forecasting the U.S. Treasury yield curve: A distributionally robust machine learning approach. *arXiv preprint arXiv:2601.04608*.

Peters, J., Bühlmann, P., and Meinshausen, N. (2016). Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5):947–1012.

Prinster, D., Han, X., Liu, A., and Saria, S. (2025). WATCH: Adaptive monitoring for AI deployments via weighted-conformal martingales. *arXiv preprint arXiv:2505.04608*.

Rothenhäusler, D., Meinshausen, N., Bühlmann, P., and Peters, J. (2021). Anchor regression: Heterogeneous data meet causality. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(2):215–246.

Shen, X., Bühlmann, P., and Taeb, A. (2023). Distributionally robust prediction with invariant gradients. *arXiv preprint arXiv:2306.01717*.

Uziel, G. and El-Yaniv, R. (2018). Growth-optimal portfolio selection under CVaR constraints. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 84 of *PMLR*, pages 48–57.

Wang, T. (2026). Online randomized distributionally robust forecast combination for dependent data. *Journal of Time Series Analysis*.

Wu, T., Pan, J., Ong, L., and Xue, B. (2026). Cost-aware adaptive conformal inference for runtime assurance in dynamic environments. *arXiv preprint arXiv:2605.24463*.

Zhang, L., Lu, S., and Zhou, Z.-H. (2018). Adaptive online learning in dynamic environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 1323–1333.

Zinkevich, M. (2003). Online convex programming and generalized infinitesimal gradient ascent. In *Proceedings of the Twentieth International Conference on Machine Learning (ICML)*, pages 928–936.